

Inflammatory signaling in human Tuberculosis granulomas is spatially organized

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Evidence abstract

Granulomas are the pathological hallmark of tuberculosis (TB).

However, their function and mechanisms of formation remain poorly understood.

To understand the role of granulomas in TB, we analyzed the proteomes of granulomas from subjects with tuberculosis in an unbiased manner.

Using laser-capture microdissection, mass spectrometry and confocal microscopy, we generated detailed molecular maps of human granulomas.

We found that the centers of granulomas have a pro-inflammatory environment that is characterized by the presence of antimicrobial peptides, reactive oxygen species and pro-inflammatory eicosanoids.

Conversely, the tissue surrounding the caseum has a comparatively anti-inflammatory signature.

These findings are consistent across a set of six human subjects and in rabbits.

Although the balance between systemic pro- and anti-inflammatory signals is crucial to TB disease outcome, here we find that these signals are physically segregated within each granuloma.

From the protein and lipid snapshots of human and rabbit lesions analyzed here, we hypothesize that the pathologic response to TB is shaped by the precise anatomical localization of these inflammatory pathways during the development of the granuloma.

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